DIAGONAL PROPERTIES ARE THICKENING-INVARIANT (17-CASE TRANSFER)

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 453-468 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 470-501, the complete author proof (30 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 35 of queue_stacks-stacks-more-morphisms.json (pb=1277, stmt 453-468, proof 470-501).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-reduced-diagonal: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 410–426. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BPJ: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 501–501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09ZY: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 501–501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04XS: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 501–501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 04XB: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 501–501. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(f, f') : (\mathcal{X} \subset \mathcal{X}') \to (\mathcal{Y} \subset \mathcal{Y}')$ be a morphism of thickenings of algebraic stacks. Let $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ and $\Delta' : \mathcal{X}' \to \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}'$ be the corresponding diagonal morphisms. Then each property from the following list is satisfied by Δ if and only if it is satisfied by Δ': (a) representable by schemes, (b) affine, (c) surjective, (d) quasi-compact, (e) universally closed, (f) integral, (g) quasi-separated, (h) separated, (i) universally injective, (j) universally open, (k) locally quasi-finite, (l) finite, (m) unramified, (n) monomorphism, (o) immersion, (p) closed immersion, and (q) proper.*

Proof. Observe that

$$(\Delta, \Delta') : (\mathcal{X} \subset \mathcal{X}') \longrightarrow (\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \subset \mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}')$$

is a morphism of thickenings (Lemma lemma-reduced-diagonal). Moreover Δ and Δ' are representable by algebraic spaces by Morphisms of Stacks, Lemma 04XS. Hence, via the discussion in Properties of Stacks, Section 04XB the lemma follows for cases (a), (b), (c), (d), (e), (f), (g), (h), (i), and (j) by using More on Morphisms of Spaces, Lemma 09ZY.

Lemma lemma-reduced-diagonal tells us that $\mathcal{X} = (\mathcal{X} \times_{\mathcal{Y}} \mathcal{X}) \times_{(\mathcal{X}' \times_{\mathcal{Y}'} \mathcal{X}')} \mathcal{X}'$. Moreover, Δ and Δ' are locally of finite type by the aforementioned Morphisms of Stacks, Lemma 04XS. Hence the result for cases (k), (l), (m), (n), (o), (p), and (q) by using More on Morphisms of Spaces, Lemma 0BPJ. □